

Feynman-Kac PDEs for Constrained and Cost Guided Diffusion

Abstract

We essentially want to guide a diffusion generative model using a collection of user-specified cost functions and constraints, with the goal of biasing samples toward low-cost, feasible regions while still remaining consistent with the underlying diffusion dynamics. The main takeaway is that Feynman–Kac formulas naturally define time-dependent *weights* that describe how the guidance should affect the sampling distribution. These weights can then be used to *correct* the guidance, so the resulting sampler matches the intended guided behavior. In this note, we derive the PDEs that produce these Feynman–Kac weights.

1 Introduction (mainly to set up notation)

Generative modeling via diffusion models can be formulated as the simulation of the Stochastic Differential Equation (SDE) corresponding to the reverse-time process. In particular, during training, one gradually destroys samples from the data-distribution $p_{\text{data}}(x)$ by simulating the following noising SDE:

$$dx_\tau = f_\tau(x_\tau)d\tau + \sigma_\tau d\bar{W}_\tau, \quad x_{\tau=0} \sim p_{\text{data}}(x), \quad (1)$$

where $f_\tau(x_\tau)$ is usually some linear drift function $f_\tau(x_\tau) = \alpha_\tau x_\tau$, σ_τ defines the scale of noise through time, and $d\bar{W}_\tau$ is the standard Wiener process. The drift f_τ and the diffusion coefficient σ_τ are chosen so the final density is close to the standard normal distribution $p_{\tau=1} \approx \mathcal{N}(0, I_d)$.

The generation process then can be defined as the family of denoising SDEs in the opposite time direction ($t = 1 - \tau$),

$$dx_t = (-f_t(x_t) + \sigma_t^2 \nabla \log p_t(x_t))dt + \sigma_t dW_t, \quad (2)$$

where $p_t = p_{1-\tau}$ is the density of the marginals induced by the noising process in [Eq. \(1\)](#); hence, the process starts with $x_0 \sim \mathcal{N}(x|0, I_d)$.

2 Background

I am going to assume that you have read the Feynman-Kac correctors paper. But for your quick reference, I will list the key (and some well known) equations which were used later.

Continuity Equation. describes the density change when the samples move in space according to a flow v_t

$$dx_t = v_t(x_t)dt \implies \frac{\partial p_t(x)}{\partial t} = -\langle \nabla, p_t(x)v_t(x) \rangle. \quad (3)$$

Diffusion Equation. describes the change of density for the pure Brownian motion with coefficient σ_t ,

$$dx_t = \sigma_t dW_t \implies \frac{\partial p_t(x)}{\partial t} = \frac{\sigma_t^2}{2} \Delta p_t(x). \quad (4)$$

Fokker-Planck Equation. The SDE in [Eq. \(2\)](#) can be viewed as the composition of a flow and diffusion terms, where the corresponding Fokker-Planck PDE describes the combined evolution

$$\frac{\partial p_t(x)}{\partial t} = -\langle \nabla, p_t(x)v_t(x) \rangle + \frac{\sigma_t^2}{2} \Delta p_t^{\text{sde}}(x). \quad (5)$$

Reweighting Equation. describes the change of density when samples have time-dependent log-weights w_t which are updated based on the positions of samples x_t ,

$$dw_t = \bar{g}_t(x_t)dt \implies \frac{\partial p_t^w(x)}{\partial t} = \bar{g}_t(x)p_t^w(x), \quad (6)$$

where $\bar{g}_t(x) = g_t(x) - \int g_t(x)p_t^w(x)dx$

where the last equation guarantees the conservation of the normalization constant, i.e. $\int dx \bar{g}_t(x)p_t^w(x) = 0$.

Feynman-Kac Formula. We now focus on the combination of all three components to describe the *Feynman-Kac PDE*,

$$\begin{aligned} \frac{\partial p_t^{\text{FK}}(x)}{\partial t} = & -\langle \nabla, p_t^{\text{FK}}(x)v_t(x) \rangle + \frac{\sigma_t^2}{2} \Delta p_t^{\text{FK}}(x) \\ & + \bar{g}_t(x)p_t^{\text{FK}}(x), \end{aligned} \quad (7)$$

where to sample from $p_t^{\text{FK}}(x)$, one first has to sample x_t via the following SDE

$$dx_t = v_t(x_t)dt + \sigma_t dW_t, \quad dw_t = \bar{g}_t(x_t)dt, \quad (8)$$

and then reweight the obtained samples using w_t . Thus, $p_t^{\text{FK}}(x)$ reflects the density of *weighted* samples, which differs from the density $p_t(x)$ obtained via the Fokker-Planck PDE in [Eq. \(5\)](#) due to the addition of reweighting terms.¹

2.1 Duality(-ish) of these Equations

Diffusion $\rightarrow$ Continuity Through simple manipulations, we can rewrite the diffusion equation using a continuity equation and change the simulation scheme accordingly

$$\begin{aligned} \frac{\partial p_t(x)}{\partial t} &= \frac{\sigma_t^2}{2} \Delta p_t(x) = -\left\langle \nabla, p_t(x) \left(-\frac{\sigma_t^2}{2} \nabla \log p_t(x) \right) \right\rangle \\ \implies dx_t &= -\frac{\sigma_t^2}{2} \nabla \log p_t(x_t)dt. \end{aligned} \quad (9)$$

The reweighting equation adds an extra dimension to the interplay between different simulation schemes.

Continuity $\rightarrow$ Reweighting We first recast the continuity equation in terms of reweighting, in which case the simulation changes the density solely by adjusting the weights of samples (without transport),

$$\begin{aligned} \frac{\partial p_t(x)}{\partial t} &= -\langle \nabla, p_t(x)v_t(x) \rangle = \left(\frac{-1}{p_t(x)} \langle \nabla, p_t(x)v_t(x) \rangle \right) p_t(x) \\ \implies dw_t &= (-\langle \nabla, v_t(x_t) \rangle - \langle \nabla \log p_t(x_t), v_t(x_t) \rangle)dt \end{aligned} \quad (10)$$

Diffusion $\rightarrow$ Reweighting We further observe that diffusion terms may be captured in the weights using

$$\begin{aligned} \frac{\partial p_t(x)}{\partial t} &= \frac{\sigma_t^2}{2} \Delta p_t(x) = \frac{\sigma_t^2}{2} p_t(x) (\Delta \log p_t(x) + \|\nabla \log p_t(x)\|^2) \\ \implies dw_t &= \frac{\sigma_t^2}{2} (\Delta \log p_t(x_t) + \|\nabla \log p_t(x_t)\|^2) dt \end{aligned} \quad (11)$$

In particular, using [Eqs. \(10\)](#) and [\(11\)](#) we now have an approach for translating arbitrary flow v_t or diffusion σ_t terms into the reweighting factors, assuming access to an exact score function $\nabla \log p_t$.

¹The Feynman-Kac correctors paper describes a couple of practical methods to sample according these weights, provide that you have the right Feynman-Kac PDEs.

3 Problem Statement

During inference (i.e., the reverse-time diffusion process), we seek to guide sampling so that the generated samples satisfy the constrained optimization problem

$$\begin{aligned} & \underset{x \in \mathcal{X}}{\text{minimize}} \quad \mathcal{L}(x) \\ & \text{subject to} \quad h_1(x) = 0, \quad h_2(x) \leq 0. \end{aligned} \quad (12)$$

A natural approach is to perform guided sampling from the time- t marginals

$$p_t(x) = \frac{1}{Z_t} q_t(x) m_t(x), \quad (13)$$

where q_t denotes the diffusion model marginal at time t , Z_t is a normalizing constant, and m_t is a nonnegative weighting function that encodes preference for feasibility and optimality over $\mathcal{X}$. In particular, $m_t(x)$ attains its maximum value on the set of optimal solutions to equation 12 and decays as x becomes increasingly suboptimal and/or violates the constraints.

Although it is possible to derive the Feynman–Kac PDEs (and hence the corresponding correction terms) for p'_t , we instead can adopt a simpler (yet equivalent) formulation. We can introduce a binary indicator $O : \mathcal{X} \rightarrow \{0, 1\}$, which flags samples that satisfy the desired optimality criterion. We define

$$\mathcal{J}(x) := \mathcal{L}(x) + \rho(x) + c_1 h_1(x)^2 + c_2 \text{softmax}\{0, h_2(x)\}^2 \quad (14)$$

The likelihood of observing $O = 1$ given a trajectory x is defined as

$$p(O = 1 \mid x) = e^{(\beta \mathcal{J}(x))}, \quad (15)$$

where $\beta < 0$ controls the strength of the preference for low-cost samples. Combining this likelihood with a prior $p(x)$ over trajectories via Bayes’ rule yields the posterior distribution

$$p(x \mid O = 1) \propto p(x) p(O = 1 \mid x). \quad (16)$$

Now our goal is to sample from the time- t marginals

$$p_t(x) := q_t(x) p_t(O = 1 \mid x) = q_t(x) e^{(\beta \mathcal{J}(x))}$$

4 Guiding Diffusion with Feynman-Kac PDEs

First, we consider guiding the diffusion model using a cost function(al) $\mathcal{J}(\cdot)$. At time t , the diffusion model produces samples from a target distribution q_t , but our goal is to instead sample from a modified distribution in which low-cost (as measured by $\mathcal{J}$) samples are assigned higher probability. Accordingly, we define the guided marginal distribution

$$p_t(x) \propto q_t(x) e^{(\beta_t \mathcal{J}(x))}, \quad (17)$$

where $\beta < 0$ controls the strength of the guidance. (This is called Gibbs tilting in statistical mechanics and thermodynamics.)

²Here $\text{softmax}\{0, h_2(x)\}$ is just a C^2 approximation of $\max\{0, h_2(x)\}$. We will use softmax over max just so that the math goes through, but in practice we can probably get away with using max.

Proposition 4.1 (Cost guided SDE). *Consider the following SDE*

$$dx_t = v_t(x)dt + \sigma_t dW_t, \quad (18)$$

which samples from the marginals $q_t(x)$. The samples from the marginals $p_t(x) \propto q_t(x)e^{(\beta_t \mathcal{J}(x))}$ can be simulated according to the following SDE

$$dx_t = (v_t(x_t) + a \nabla \mathcal{J}(x_t))dt + \sigma_t dW_t, \quad (19)$$

$$dw_t = \left[\left\langle \nabla \mathcal{J}(x_t), \beta_t(v_t(x_t) - \sigma_t^2 \nabla \log q_t(x_t) - \frac{\sigma_t^2}{2} \beta_t \nabla \mathcal{J}(x_t)) + a(\nabla \log q_t(x_t) + \beta_t \nabla \mathcal{J}(x_t)) \right\rangle + \right. \quad (20)$$

$$\left. + \left(a - \beta_t \frac{\sigma_t^2}{2} \right) \Delta \mathcal{J}(x_t) + \frac{\partial \beta_t}{\partial t} \mathcal{J}(x_t) \right] dt. \quad (21)$$

For the reverse-time SDE with drift $v_t(x_t) = -f_t(x_t) + \sigma_t^2 \nabla \log q_t(x_t)$ corresponding to the original diffusion generative model and $a = \beta_t \sigma_t^2 / 2$, we obtain the following weighted SDE

$$dx_t = (-f_t(x_t) + \sigma_t^2 \nabla \log q_t(x_t) + \beta_t \frac{\sigma_t^2}{2} \nabla \mathcal{J}(x_t))dt + \sigma_t dW_t, \quad (22)$$

$$dw_t = \left[\frac{\partial \beta_t}{\partial t} \mathcal{J}(x) + \left\langle \beta_t \nabla \mathcal{J}(x), \frac{\sigma_t^2}{2} \nabla \log q_t(x) - f_t(x) \right\rangle \right] dt \quad (23)$$

Proof. First, consider the density $q_t(x)$ that follows the PDE

$$\frac{\partial q_t(x)}{\partial t} = -\langle \nabla, q_t(x) v_t(x) \rangle + \frac{\sigma_t^2}{2} \Delta q_t(x). \quad (24)$$

We want to find the PDE for the cost guided density

$$p_t(x) = \frac{q_t(x) e^{(\beta_t \mathcal{J}(x))}}{\int dx q_t(x) e^{(\beta_t \mathcal{J}(x))}}. \quad (25)$$

Clearly, we get

$$\frac{\partial}{\partial t} \log p_t(x) = \frac{\partial}{\partial t} \log q_t(x) + \frac{\partial \beta_t}{\partial t} \mathcal{J}(x) - \int dx p_t(x) \left[\frac{\partial}{\partial t} \log q_t(x) + \frac{\partial \beta_t}{\partial t} \mathcal{J}(x) \right] \quad (26)$$

Notice that for the first term $(\frac{\partial}{\partial t} \log q_t(x))$, can be rewritten as

$$\frac{\partial}{\partial t} \log q_t(x) = -\langle \nabla, v_t(x) \rangle - \langle \nabla \log q_t(x), v_t(x) \rangle + \frac{\sigma_t^2}{2} \Delta \log q_t(x) + \frac{\sigma_t^2}{2} \|\nabla \log q_t(x)\|^2 \quad (27)$$

$$= -\langle \nabla, v_t(x) \rangle - \langle \nabla \log p_t(x), v_t(x) \rangle + \frac{\sigma_t^2}{2} \Delta \log p_t(x) + \frac{\sigma_t^2}{2} \|\nabla \log p_t(x)\|^2 + \quad (28)$$

$$+ \left\langle \beta_t \nabla \mathcal{J}(x), v_t(x) - \sigma_t^2 \nabla \log q_t(x) - \frac{\sigma_t^2}{2} \beta_t \nabla \mathcal{J}(x) \right\rangle - \beta_t \frac{\sigma_t^2}{2} \Delta \mathcal{J}(x).$$

Hence, we have the following

$$\frac{\partial p_t(x)}{\partial t} = -\langle \nabla, p_t(x) v_t(x) \rangle + \frac{\sigma_t^2}{2} \Delta p_t(x) + p_t(x) (g_t(x) - \mathbb{E}_{p_t(x)} g_t(x)) \quad (29)$$

$$g_t(x) = \left\langle \beta_t \nabla \mathcal{J}(x), v_t(x) - \sigma_t^2 \nabla \log q_t(x) - \frac{\sigma_t^2}{2} \beta_t \nabla \mathcal{J}(x) \right\rangle - \beta_t \frac{\sigma_t^2}{2} \Delta \mathcal{J}(x) + \frac{\partial \beta_t}{\partial t} \mathcal{J}(x). \quad (30)$$

Now, we can introduce the gradient of the cost as an additional drift term $a \nabla \mathcal{J}(x)$, i.e.

$$\frac{\partial p_t(x)}{\partial t} = -\langle \nabla, p_t(x) (v_t(x) + a \nabla \mathcal{J}(x)) \rangle + \frac{\sigma_t^2}{2} \Delta p_t(x) + p_t(x) (g_t(x) - \mathbb{E}_{p_t(x)} g_t(x)) \quad (31)$$

$$g_t(x) = a \Delta \mathcal{J}(x) + a \langle \nabla \log p_t(x), \nabla \mathcal{J}(x) \rangle - \beta_t \frac{\sigma_t^2}{2} \Delta \mathcal{J}(x) + \frac{\partial \beta_t}{\partial t} \mathcal{J}(x) + \left\langle \beta_t \nabla \mathcal{J}(x), v_t(x) - \sigma_t^2 \nabla \log q_t(x) - \frac{\sigma_t^2}{2} \beta_t \nabla \mathcal{J}(x) \right\rangle. \quad (32)$$

By setting $v_t(x) := -f_t(x) + \sigma_t^2 \nabla \log q_t(x)$ and $a := \beta_t \sigma_t^2 / 2$, we get

$$\begin{aligned} \frac{\partial p_t(x)}{\partial t} &= - \left\langle \nabla, p_t(x) (-f_t(x) + \sigma_t^2 \nabla \log q_t(x) + \beta_t \frac{\sigma_t^2}{2} \nabla \mathcal{J}(x)) \right\rangle + \frac{\sigma_t^2}{2} \Delta p_t(x) + p_t(x) (g_t(x) - \mathbb{E}_{p_t(x)} g_t(x)) \\ g_t(x) &= \left\langle \beta_t \nabla \mathcal{J}(x), \frac{\sigma_t^2}{2} \nabla \log p_t(x) \right\rangle + \frac{\partial \beta_t}{\partial t} \mathcal{J}(x) + \left\langle \beta_t \nabla \mathcal{J}(x), -f_t(x) - \frac{\sigma_t^2}{2} \beta_t \nabla \mathcal{J}(x) \right\rangle \end{aligned} \quad (33)$$

$$= \frac{\partial \beta_t}{\partial t} \mathcal{J}(x) + \left\langle \beta_t \nabla \mathcal{J}(x), \frac{\sigma_t^2}{2} \nabla \log q_t(x) - f_t(x) \right\rangle. \quad (34)$$

This can be simulated as

$$dx_t = (-f_t(x_t) + \sigma_t^2 \nabla \log q_t(x_t) + \beta_t \frac{\sigma_t^2}{2} \nabla \mathcal{J}(x_t)) dt + \sigma_t dW_t, \quad (35)$$

$$dw_t = \left[\frac{\partial \beta_t}{\partial t} \mathcal{J}(x) + \left\langle \beta_t \nabla \mathcal{J}(x), \frac{\sigma_t^2}{2} \nabla \log q_t(x) - f_t(x) \right\rangle \right] dt \quad (36)$$

□

5 Convergence Guarantees

Proposition 5.1 (Approximate constrained optimization by Gibbs tilting + penalty growth). *Let $X \subset \mathbb{R}^d$ be a compact set with nonempty interior (or at least positive Lebesgue measure), and let $L, \rho, h_1, h_2 : X \rightarrow \mathbb{R}$ be continuous. Define the feasible set*

$$F := \{x \in X : h_1(x) = 0, h_2(x) \leq 0\} \neq \emptyset, \quad L^* := \min_{x \in F} L(x),$$

and for penalties $c = (c_1, c_2)$ with $c_1, c_2 > 0$ define the penalized objective

$$J_c(x) := L(x) + \rho(x) + c_1 h_1(x)^2 + c_2 \max\{0, h_2(x)\}.$$

Let $q_0 : X \rightarrow [0, \infty)$ be a continuous density (with respect to Lebesgue measure) and, for $\beta < 0$, define the Gibbs-tilted probability distribution

$$p^{(\beta, c)}(dx) := \frac{q_0(x) e^{\beta J_c(x)}}{\int_X q_0(u) e^{\beta J_c(u)} du} dx.$$

Assume the following:

(A1) (**Compactness**) X is compact.

(A2) (**Continuity**) J_c is continuous on X for each fixed c .

(A3) (**Positivity near minimizers**) For each fixed c , if $S_c := \arg \min_X J_c$ is the minimizer set, then there exists an open set $U_c \subset X$ with $U_c \cap S_c \neq \emptyset$ and a constant $m_c > 0$ such that $q_0(x) \geq m_c$ for all $x \in U_c$, and $\text{Leb}(U_c) > 0$.

(A4) (**Penalty correctness for global minimizers**) For every $\delta > 0$, there exists $c = (c_1, c_2)$ such that every global minimizer $x_c \in \arg \min_X J_c$ satisfies

$$|h_1(x_c)| \leq \delta/2, \quad h_2(x_c) \leq \delta/2, \quad L(x_c) \leq L^* + \delta/2.$$

Then for every $\delta > 0$ and $\eta \in (0, 1)$, there exist penalties $c = (c_1, c_2)$ (large enough) and $\beta < 0$ (with $|\beta|$ large enough) such that, for $X \sim p^{(\beta, c)}$,

$$\mathbb{P}(|h_1(X)| \leq \delta, h_2(X) \leq \delta, L(X) \leq L^* + \delta) \geq 1 - \eta.$$

Proof. Fix arbitrary $\delta > 0$ and $\eta \in (0, 1)$. Define the “good” set

$$G_\delta := \left\{x \in X : |h_1(x)| \leq \delta, h_2(x) \leq \delta, L(x) \leq L^* + \delta\right\}, \quad (37)$$

and its complement (the “bad” set)

$$A_\delta := X \setminus G_\delta.$$

Choose penalties so that all minimizers of J_c are $\delta/2$ -good. By assumption (A4), there exists a choice of penalties $c = (c_1, c_2)$ such that every global minimizer $x_c \in S_c := \arg \min_X J_c$ satisfies

$$|h_1(x_c)| \leq \delta/2, \quad h_2(x_c) \leq \delta/2, \quad L(x_c) \leq L^* + \delta/2.$$

Equivalently,

$$S_c \subseteq G_{\delta/2} \subseteq G_\delta. \quad (38)$$

Fix such a c for the remainder of the proof, and write $J := J_c$ and $S := S_c$ to simplify notation. Let

$$J^* := \min_{x \in X} J(x),$$

so that $S = \{x \in X : J(x) = J^*\}$ and $S \neq \emptyset$ by compactness (A1) and continuity (A2).

There is a strictly positive energy gap outside G_δ . We claim there exists $\varepsilon > 0$ such that

$$A_\delta \subseteq \{x \in X : J(x) \geq J^* + \varepsilon\}. \quad (39)$$

To prove this, note first that G_δ is closed in X because it is defined by finitely many inequalities of continuous functions; hence $A_\delta = X \setminus G_\delta$ is open in X . Since we will use compactness, we instead work with the closed superset

$$\tilde{A}_\delta := \left\{x \in X : |h_1(x)| \geq \delta \text{ or } h_2(x) \geq \delta \text{ or } L(x) \geq L^* + \delta\right\}.$$

Each of the sets $\{|h_1| \geq \delta\}$, $\{h_2 \geq \delta\}$, and $\{L \geq L^* + \delta\}$ is closed in X by continuity, so $\tilde{A}_\delta$ is closed as a finite union of closed sets. Moreover, by definition we have the inclusion $A_\delta \subseteq \tilde{A}_\delta$. Since X is compact, it follows that $\tilde{A}_\delta$ is compact.

Now define the continuous nonnegative function

$$\Delta(x) := J(x) - J^* \geq 0.$$

Since $\tilde{A}_\delta$ is compact and Δ is continuous, Δ attains its minimum on $\tilde{A}_\delta$:

$$\varepsilon := \min_{x \in \tilde{A}_\delta} \Delta(x) = \min_{x \in \tilde{A}_\delta} (J(x) - J^*).$$

We show $\varepsilon > 0$. Suppose for contradiction that $\varepsilon = 0$. Then there exists $x_0 \in \tilde{A}_\delta$ with $J(x_0) - J^* = 0$, i.e. $J(x_0) = J^*$, so $x_0 \in S$. But by equation 38, $S \subseteq G_{\delta/2}$, and hence $|h_1(x_0)| \leq \delta/2$, $h_2(x_0) \leq \delta/2$, and $L(x_0) \leq L^* + \delta/2$, which implies $x_0 \notin \tilde{A}_\delta$. This is a contradiction. Therefore $\varepsilon > 0$, and by definition of ε we have

$$J(x) - J^* \geq \varepsilon \quad \text{for all } x \in \tilde{A}_\delta.$$

Using $A_\delta \subseteq \tilde{A}_\delta$, we conclude that

$$J(x) - J^* \geq \varepsilon \quad \text{for all } x \in A_\delta,$$

which is exactly equation 39.

Consequently,

$$p^{(\beta, c)}(A_\delta) \leq p^{(\beta, c)}(\{x : J(x) \geq J^* + \varepsilon\}) \quad \text{for every } \beta < 0. \quad (40)$$

Exponential bound on the energy tail as $\beta \rightarrow -\infty$. Define the “high-energy” set

$$B := \{x \in X : J(x) \geq J^* + \varepsilon\}.$$

We will show that $p^{(\beta, c)}(B) \rightarrow 0$ as $\beta \rightarrow -\infty$, and in fact we will obtain an explicit bound.

By (A3), there exists an open set $U \subset X$ with $\text{Leb}(U) > 0$ and constants $m > 0$ such that: (i) $U \cap S \neq \emptyset$, and (ii) $q_1(x) \geq m$ for all $x \in U$.

Pick any $x^* \in U \cap S$. Since x^* is a minimizer, $J(x^*) = J^*$. By continuity of J , there exists a (possibly smaller) open neighborhood $U' \subseteq U$ of x^* such that

$$J(x) \leq J^* + \varepsilon/2 \quad \text{for all } x \in U'. \quad (41)$$

Because $U' \subseteq U$ and $q_1 \geq m$ on U , we also have

$$q_0(x) \geq m \quad \text{for all } x \in U', \quad \text{and} \quad \text{Leb}(U') > 0. \quad (42)$$

Also, since q_1 is continuous on compact X , it is bounded above:

$$\exists M < \infty \quad \text{such that} \quad q_0(x) \leq M \quad \forall x \in X. \quad (43)$$

Now compute, for $\beta < 0$,

$$p^{(\beta, c)}(B) = \frac{\int_B q_0(x) e^{\beta J(x)} dx}{\int_X q_0(x) e^{\beta J(x)} dx}. \quad (44)$$

Upper bound the numerator. For all $x \in B$, $J(x) \geq J^* + \varepsilon$. Since $\beta < 0$, the function $t \mapsto e^{\beta t}$ is decreasing, so

$$e^{\beta J(x)} \leq e^{\beta(J^* + \varepsilon)} \quad \forall x \in B.$$

Using equation 43,

$$\int_B q_0(x) e^{\beta J(x)} dx \leq \int_B M e^{\beta(J^* + \varepsilon)} dx = M \text{Leb}(B) e^{\beta(J^* + \varepsilon)} \leq M \text{Leb}(X) e^{\beta(J^* + \varepsilon)}.$$

Thus

$$\int_B q_0(x) e^{\beta J(x)} dx \leq M \text{Leb}(X) e^{\beta(J^* + \varepsilon)}. \quad (45)$$

Lower bound the denominator. Restrict the denominator integral to the set U' . For all $x \in U'$, by equation 41 we have $J(x) \leq J^* + \varepsilon/2$, and again since $\beta < 0$,

$$e^{\beta J(x)} \geq e^{\beta(J^* + \varepsilon/2)} \quad \forall x \in U'.$$

Using equation 42,

$$\int_X q_0(x) e^{\beta J(x)} dx \geq \int_{U'} q_0(x) e^{\beta J(x)} dx \geq \int_{U'} m e^{\beta(J^* + \varepsilon/2)} dx = m \text{Leb}(U') e^{\beta(J^* + \varepsilon/2)}.$$

Hence

$$\int_X q_0(x) e^{\beta J(x)} dx \geq m \text{Leb}(U') e^{\beta(J^* + \varepsilon/2)}. \quad (46)$$

Combine bounds. Substituting equation 45 and equation 46 into equation 44 yields

$$p^{(\beta, c)}(B) \leq \frac{M \text{Leb}(X) e^{\beta(J^* + \varepsilon)}}{m \text{Leb}(U') e^{\beta(J^* + \varepsilon/2)}} = \frac{M \text{Leb}(X)}{m \text{Leb}(U')} e^{\beta \varepsilon/2}.$$

Therefore, for all $\beta < 0$,

$$p^{(\beta, c)}(B) \leq C e^{\beta \varepsilon/2}, \quad \text{where } C := \frac{M \text{Leb}(X)}{m \text{Leb}(U')} \in (0, \infty). \quad (47)$$

Since $\varepsilon > 0$ and $\beta \rightarrow -\infty$, we have $e^{\beta \varepsilon/2} \rightarrow 0$, hence $p^{(\beta, c)}(B) \rightarrow 0$. In particular, choose $\beta < 0$ sufficiently negative so that

$$C e^{\beta \varepsilon/2} \leq \eta. \quad (48)$$

For example, it suffices to take any $\beta < 0$ satisfying

$$\beta \leq \frac{2}{\varepsilon} \log\left(\frac{\eta}{C}\right) \quad (\text{note: the right-hand side is negative because } \eta/C < 1 \text{ for large enough } |\beta|).$$

Conclude the desired high-probability guarantee on G_δ . By equation 40 and equation 48,

$$p^{(\beta, c)}(A_\delta) \leq p^{(\beta, c)}(B) \leq \eta.$$

Therefore,

$$p^{(\beta, c)}(G_\delta) = 1 - p^{(\beta, c)}(A_\delta) \geq 1 - \eta.$$

By the definition equation 37 of G_δ , this is exactly

$$\mathbb{P}\left(|h_1(X)| \leq \delta, \quad h_2(X) \leq \delta, \quad L(X) \leq L^* + \delta\right) \geq 1 - \eta, \quad \text{for } X \sim p^{(\beta, c)}.$$

□

Corollary 5.2 (Explicit success/failure bound implied by Cor. 5.1). *For a fixed $\delta > 0$ and any target failure level $\eta \in (0, 1)$, it suffices to choose*

$$\beta \leq \frac{2}{\varepsilon(c, \delta)} \log\left(\frac{\eta}{C(c, \delta)}\right) \implies \mathbb{P}_{p_{\beta, c}}(x \in A_\delta) \leq \eta.$$

Equivalently,

$$\mathbb{P}_{p_{\beta, c}}(x \in G_\delta) \geq 1 - C(c, \delta) \exp\left(\frac{\beta \varepsilon(c, \delta)}{2}\right).$$

In essence, β tunes how strongly we weight constraint satisfaction and cost minimization relative to the baseline (target/data) distribution. I am sorry, I do not have a good closed form bound for β that is useful in practice. But we can get much stronger claims and grantees if we take constraints out of the picture.